

Breaking a Weak RSA Key

Public Key: $(n, e) = (1349, 527)$

Introduction

Bob is using a weak RSA key with modulus $n = 1349$, small enough to factor by hand. Our goals are:

- (a) Factor n , compute $\varphi(n)$, and recover Bob's private key d .
- (b) Decrypt the ciphertext $M = 1209$ to find the plaintext m .

(a) Breaking Bob's RSA Key

Step 1: Factor the modulus n

We attempt small primes and find:

$$1349 = 19 \times 71.$$

Therefore,

$$p = 19, \quad q = 71.$$

Step 2: Compute Euler's Totient

$$\varphi(n) = (p - 1)(q - 1) = 18 \cdot 70 = 1260.$$

Step 3: Find the private exponent d

We need d such that

$$527d \equiv 1 \pmod{1260}.$$

Thus d is the modular inverse of 527 modulo 1260.

Step 3a: Euclidean Algorithm

$$1260 = 2 \cdot 527 + 206$$

$$527 = 2 \cdot 206 + 115$$

$$206 = 1 \cdot 115 + 91$$

$$115 = 1 \cdot 91 + 24$$

$$91 = 3 \cdot 24 + 19$$

$$24 = 1 \cdot 19 + 5$$

$$19 = 3 \cdot 5 + 4$$

$$5 = 1 \cdot 4 + 1$$

$$4 = 4 \cdot 1 + 0$$

Thus $\gcd(527, 1260) = 1$ and an inverse exists.

Step 3b: Extended Euclidean Algorithm

We back-substitute to express 1 as a linear combination:

$$\begin{aligned}1 &= 5 - 4 \\4 &= 19 - 3 \cdot 5 \\1 &= 4 \cdot 5 - 19 \\5 &= 24 - 19 \\1 &= 4 \cdot 24 - 5 \cdot 19 \\19 &= 91 - 3 \cdot 24 \\1 &= 19 \cdot 24 - 5 \cdot 91 \\24 &= 115 - 91 \\1 &= 19 \cdot 115 - 24 \cdot 91 \\91 &= 206 - 115 \\1 &= 43 \cdot 115 - 24 \cdot 206 \\115 &= 527 - 2 \cdot 206 \\1 &= 43 \cdot 527 - 110 \cdot 206 \\206 &= 1260 - 2 \cdot 527 \\1 &= 263 \cdot 527 - 110 \cdot 1260.\end{aligned}$$

Thus,

$$\boxed{d = 263}.$$

Bob's private key is therefore:

$$\boxed{(n, d) = (1349, 263)}.$$

(b) Decrypt the Ciphertext

Alice sends the encrypted message:

$$M = 1209.$$

To decrypt, compute:

$$m = M^d \bmod n = 1209^{263} \bmod 1349.$$

We use the Chinese Remainder Theorem with the factors $p = 19$ and $q = 71$.

Step 1: Reduce Modulo $p = 19$

$$1209 \equiv 12 \pmod{19}.$$

We use Fermat's theorem ($a^{18} \equiv 1 \pmod{19}$):

$$263 = 18 \cdot 14 + 11 \Rightarrow 263 \equiv 11 \pmod{19}.$$

Thus

$$m \equiv 12^{11} \pmod{19}.$$

Compute by repeated squaring:

$$12^2 \equiv 11, \quad 12^4 \equiv 7, \quad 12^8 \equiv 11.$$

As $11 = 8 + 2 + 1$:

$$12^{11} \equiv 11 \cdot 11 \cdot 12 \equiv 8 \pmod{19}.$$

So:

$$m \equiv 8 \pmod{19}.$$

Step 2: Reduce Modulo $q = 71$

$$1209 \equiv 2 \pmod{71}.$$

Fermat's theorem ($2^{70} \equiv 1 \pmod{71}$):

$$263 = 70 \cdot 3 + 53 \Rightarrow 2^{263} \equiv 2^{53} \pmod{71}.$$

Repeated squaring gives:

$$2^{53} \equiv 12 \pmod{71}.$$

Thus:

$$m \equiv 12 \pmod{71}.$$

Step 3: Combine Using CRT

We solve:

$$\begin{cases} m \equiv 8 \pmod{19}, \\ m \equiv 12 \pmod{71}. \end{cases}$$

Let:

$$m = 8 + 19k.$$

Insert into the second congruence:

$$8 + 19k \equiv 12 \pmod{71}$$

$$19k \equiv 4 \pmod{71}.$$

The inverse of 19 modulo 71 is 15 (since $19 \cdot 15 \equiv 1 \pmod{71}$).

Thus:

$$k \equiv 4 \cdot 15 = 60 \pmod{71}.$$

Use $k = 60$:

$$m = 8 + 19 \cdot 60 = 1148.$$

Therefore the plaintext is:

$$\boxed{m = 1148}.$$

Final Answers

Private key: $d = 263$

Plaintext: $m = 1148$
